



Lenacapavir for HIV PrEP: Narrative Review of Clinical Trials and Implications for Patient Access

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Abstract

Purpose of Review By the end of 2023, approximately 39.9 million people worldwide were living with HIV, including 1.2 million in the United States. Although treatment advancements have improved health outcomes, HIV continues to pose significant challenges, particularly among key populations like men who have sex with men (MSM), transgender women, and cisgender women. Pre-exposure prophylaxis (PrEP) has emerged as an effective strategy to lower the risk of HIV transmission, but daily oral regimens often face adherence difficulties. This narrative review aims to summarize the effectiveness and safety of Lenacapavir, the latest long-acting injectable PrEP currently under clinical trials.

Recent Findings Lenacapavir, a novel long-acting injectable PrEP with bi-annual dosing, offers significant advantages in terms of adherence and privacy, potentially reducing the stigma associated with daily medication. This review summarizes the effectiveness in preventing HIV, safety profile and the strategies being explored to overcome barriers to access. Compared to traditional daily oral PrEP options, lenacapavir has shown remarkable efficacy in recent clinical trials, including two different Phase 3 studies where it achieved a 100% prevention rate in cisgender women and a 96% prevention rate in cisgender men who have sex with men, transgender men, transgender women and gender non-binary individuals.

Summary Lenacapavir holds promise as a transformative option in HIV prevention. However, continued research and development of strategies to improve access are essential to meet public health objectives and ultimately end the HIV epidemic by 2030.

Keywords HIV PrEP · Lenacapavir · Efficacy · Clinical trials · Accessibility

Background

Globally, around 39.9 million people were living with human immunodeficiency virus (HIV) by the end of 2023, with 1.2 million cases reported in the United States alone[1]. Despite the advancements in treatment that have improved outcomes for many, HIV remains a critical issue, especially among key populations like men who have sex with men (MSM), transgender women and people who inject drugs and cisgender women.[2, 3]. Pre-exposure prophylaxis (PrEP) has emerged as one of the most effective strategies to prevent HIV [4]. Oral PrEP options, such as tenofovir

disoproxil fumarate/emtricitabine (TDF/FTC) and tenofovir alafenamide/emtricitabine (TAF/FTC), are highly effective, but these daily regimens are associated with challenges related to adherence[5]. Studies have shown that missing doses of daily PrEP can significantly reduce its effectiveness in preventing HIV[6].

To overcome the challenges of adherence, long acting PrEP such as cabotegravir which is a bi-monthly injection approved in 2021[7] and lenacapavir which is a twice yearly injectable under development, are promising tools. Lenacapavir received European Commission approval on August 22, 2022, and FDA approval on December 22, 2022, for the treatment of adults with multidrug-resistant HIV-1 infections. Lenacapavir is being studied as an investigational drug for PrEP and Gilead Sciences recently announced promising interim findings from the Phase 3 PURPOSE 1 clinical trial, showcasing the efficacy of lenacapavir[8]. The trial, which focused on cisgender women—a group disproportionately affected by HIV in certain regions—reported a 100% prevention rate of

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HIV-1 infections among participants receiving lenacapavir[8]. This was a notable improvement over the once-daily oral PrEP medication, TDF/FTC which saw a higher incidence of infections in the same population[8]. Lenacapavir offers the advantage of reduced the frequency of dosing, potentially improve adherence among individuals who face challenges with daily medication due to time constraints, privacy concerns, or the stigma associated with PrEP use[9].

Pharmacology of Lenacapavir

Lenacapavir is a small molecule with a molecular formula of C₃₉H₃₂ClF₁₀N₇O₅S₂ and a molecular weight of approximately 968.28 Da. Lenacapavir is a capsid inhibitor, which means it interferes with the HIV-1 capsid, a crucial protein shell surrounding the viral RNA and enzymes. The drug's mechanism of action involves binding to and disrupting the HIV-1 capsid, thereby preventing viral replication by inhibiting the proper assembly and function of the viral core[10]. The pharmacokinetics of Lenacapavir are characterized by its extended half-life, supporting its use as a long-acting PrEP regimen. Lenacapavir is available in both oral and subcutaneous injection forms[11]. The subcutaneous route provides a slow, sustained release, offering the potential for long-acting dosing. The median half-life ranged from 10 to 12 days following oral administration, and 8 to 12 weeks following subcutaneous administration. It is slowly released, achieving peak plasma concentrations approximately 84 days post-dose[12]. Oral bioavailability is relatively low (6–10%), but lenacapavir maintains effective plasma concentrations for extended periods, making it suitable for long-acting treatment regimens. It is primarily metabolized by cytochrome P450 enzymes, particularly CYP3A. This metabolism suggests potential interactions with other drugs that inhibit or induce CYP3A. Drug interactions can impact lenacapavir's effectiveness. For example, it can elevate serum concentrations of medications such as abrocitinib, and its metabolism may be influenced by agents like abemaciclib, requiring careful consideration of co-administered drugs.

Despite the promising pharmacology and mode of administration of lenacapavir as PrEP, challenges such as affordability and equitable access persist. This article delves into its effectiveness in preventing HIV, safety profile and the strategies being explored to overcome barriers to access, providing a detailed look at how lenacapavir will influence the landscape of HIV prevention.

Methods

A literature search for the term “lenacapavir” along with “pre-exposure prophylaxis” and “HIV” was performed using PubMed. The search was conducted for studies published

prior to September 2024. No date filter was applied to the search start date and all relevant studies were included. Clinical studies (Phase I-III) available as full text and in English were reviewed. Safety endpoints such as serious adverse drug events and efficacy outcomes such as HIV reduction were summarized into a narrative review.

Results

Efficacy of Lenacapavir

It is important to note that both Phase Ib and Phase II trials for lenacapavir took place exclusively among HIV positive participants and were focused on assessing its safety and efficacy in the treatment of HIV. In phase Ib trial, lenacapavir when compared with placebo showed significant ($p < 0.0001$) reduction in HIV-1 RNA levels [13] while in the Phase II trial CALIBRATE, treatment naïve people living with HIV-1 achieved high rates of viral suppression in 92–100% of the participants by week 28 and sustained through week 54 among 85–92% of the participants[14]. Lenacapavir was generally well-tolerated, with only minor side effects, such as injection site reactions, reported by participants. No significant safety concerns were observed during the trial, marking lenacapavir as a safe and highly effective option in HIV treatment.

Additionally, Gilead's ongoing PURPOSE program, is aimed at further investigating lenacapavir's potential as long acting PrEP, across diverse populations, including MSM and transgender women[15]. PURPOSE 1 study was a Phase III study to evaluate the use of twice yearly lenacapavir as PrEP among women at risk of HIV. It demonstrated a 100% reduction in HIV incidence with lenacapavir[16]. Lenacapavir as PrEP significantly outperformed both background HIV incidence (incidence rate ratio, 0.00; $P < 0.001$) and TDF/FTC group ($n = 1,068$; incidence rate ratio, 0.00; $P < 0.001$)[16]. Among 2,136 participants receiving TAF/FTC, there was an incidence of 2.02 per 100 person-years and this was not significantly different from the background rate. The study found that adherence strongly impacted efficacy, particularly in the TAF/FTC group, where participants with higher adherence had a significantly lower risk of HIV infection. Overall, lenacapavir was the most effective, while the efficacy of TAF/FTC and TDF/FTC was reduced due to low adherence to the daily regimen[16].

PURPOSE 2 study was a Phase III study to evaluate the use of twice yearly lenacapavir as PrEP among 3,273 cis-gender men who have sex with men, transgender people and non-binary individuals who have sex with male partners[17]. In this study there were 2 incident cases among 2,180 participants in the lenacapavir group (incidence 0.10 per 100 person-years); 99.9% of participants did not acquire HIV in

the lenacapavir group[17]. The results demonstrated superiority of twice-yearly lenacapavir over background HIV (incidence 2.37 per 100 person-years), with 96% relative risk reduction (incidence rate ratio 0.04, $p < 0.0001$) when compared with the TDF/FTC group[17] compared with 9 incident cases among 1,087 individuals in the TDF/FTC group (incidence 0.93 per 100 person-years). Twice-yearly lenacapavir was 89% more effective than once daily TDF/FTC (incidence rate ratio 0.11, $p = 0.00245$)[17]. Efficacy results from PURPOSE 1 and PURPOSE 2 are summarized in Table 1.

Safety of Lenacapavir

In phase 1b trial, 56% of lenacapavir users reported injection site reactions which were generally mild and did not lead to treatment discontinuation. One serious adverse event of Grade 3 atrial fibrillation which was deemed unrelated to lenacapavir was reported[13]. In Phase II CALIBRATE trial, 55% of lenacapavir users reported injection site reaction of which majority were classified as mild. 10 serious adverse reactions were reported and none of them were considered related to the treatment[14].

PURPOSE 1 study demonstrated that adverse events were common across all treatment groups, but most were mild or moderate in severity. The most frequent adverse events were injection-site reactions, headaches, and genitourinary infections. Of the 10,154 injections administered to 2138, 68.8% of participants who received lenacapavir experienced injection-site reactions, but these were generally mild and decreased with subsequent injections. Headaches occurred in 13.3% of the lenacapavir group, compared to 16.5% in the

TAF/FTC group and 14.5% in the TDF/FTC group[16]. Serious adverse events were relatively uncommon, reported in 2.8% of lenacapavir participants, 4.0% of TAF/FTC recipients, and 3.3% of those on TDF/FTC[16]. Few participants discontinued the trial due to adverse events, with only 0.2% in the lenacapavir group doing so. Importantly, no deaths were related to the trial drugs, and overall, the safety profile of lenacapavir was favorable, with fewer gastrointestinal side effects than TAF/FTC or TDF/FTC[16].

In PURPOSE 2, lenacapavir and TDF/FTC were generally well-tolerated, and no significant or new safety concerns were identified. Serious adverse events were observed in 3.3% of participants in the lenacapavir group, compared to 4.0% in the Truvada group[17]. The most common non-injection site adverse events were rectal chlamydia infection (13.2% lenacapavir; 11.8% TDF/FTC), oropharyngeal gonococcal infection (13.0% lenacapavir; 10.9% TDF/FTC), and rectal gonococcal infection (10.7% lenacapavir; 9.1% TDF/FTC)[17]. Most injection-site reactions were subcutaneous nodules (lenacapavir: 63.4%; placebo: 39.2%), injection site pain (lenacapavir: 56.4%; placebo: 53.4%), and erythema (lenacapavir: 17.3%; placebo: 19.4%), with fewer than 1% ($n = 29$) of participants discontinuing due to these events[17]. Safety results from PURPOSE 1 and PURPOSE 2 are summarized in Table 1.

Other ongoing studies accessing lenacapavir's PrEP capabilities are PURPOSE 3–5. PURPOSE 3 will study lenacapavir for PrEP among cisgender women in the United States who are disproportionately affected by HIV, with a focus on Black women and other women of color, and PURPOSE 4 will study lenacapavir for PrEP among people who inject drugs in the United States[18]. PURPOSE 5, the first Phase

Table 1 Summary of phase III trials for lenacapavir as PrEP

Reference	Patients	Dosing	Age	Gender	Efficacy	Safety and Adverse Events
PURPOSE 1 [16]	5345 (2134-Lenacapavir) (2136-TAF/FTC) (1068- TDF/FTC)	Lenacapavir: twice yearly TAF/FTC and TDF/FTC: daily oral tablets	16–25 years	Female	Lenacapavir: 100% reduction in HIV incidence TAF/FTC: 2.02 per 100 person-years TDF/FTC: 1.69 per 100 person-years	Lenacapavir: 76.3% with adverse events, 2.8% serious adverse events TAF/FTC: 77.9% with adverse events, 4.0% serious adverse events TDF/FTC: 77.6% with adverse events, 3.3% serious adverse events
PURPOSE 2 [17]	3,273 (2,180-Lenacapavir) (1,087-TDF/FTC)	Lenacapavir: twice yearly TDF/FTC: daily oral tablets	16+ years	Cisgender men, transgender men, transgender women, and gender non-binary individuals	Lenacapavir: 0.10 per 100 person-years TDF/FTC: 0.93 per 100 person-years bHIV: 2.37 per 100 person-years	Lenacapavir: 3.3% with serious adverse events TDF/FTC: 4.0% with serious adverse events

2 clinical trial to evaluate an investigational long-acting HIV prevention option in Europe specifically France and UK[18].

Relevance to Patient Care and Clinical Practice

The CDC put forth the ambitious goal of 90% of HIV infections in the US by 2030, of which HIV prevention was noted as one of the key strategies[19]. Approval of long-acting PrEP plays a pivotal role in helping the US make progress towards this goal. Lenacapavir being a twice-yearly PrEP regimen, reduces the adherence burden associated with daily PrEP regimens, which can be difficult for some patients to maintain. By simplifying the dosing schedule, lenacapavir can improve adherence rates, thereby enhancing its effectiveness in preventing HIV transmission among high-risk individuals, as seen in the clinical trials [20]. Additionally, the reduced frequency of dosing allows for enhanced privacy and confidentiality which was one of the key concerns noted among people at risk of HIV [21]. The reduced frequency of dosing minimizes the risk of accidental disclosure associated with daily medications. This privacy is particularly important for individuals experiencing intimate partner violence (IPV) or those who may face or anticipate stigma and discrimination related to their HIV status or PrEP use [22–24]. Lenacapavir's twice yearly dosing has the potential to improve PrEP access among individuals in areas with limited access to healthcare. By requiring fewer clinic visits, lenacapavir can serve as a more feasible option for people living in remote or underserved areas, where frequent healthcare visits are impractical or impossible. However, there are certain challenges and considerations around the uptake and implementation of long-acting injectable PrEP such as lenacapavir. Regular HIV testing is essential before each injection to ensure that the individual is HIV-negative, as undiagnosed infections could lead to the development of drug resistance [25]. Strategies to simplify and increase access to HIV testing must be integrated into any rollout plan for lenacapavir to mitigate this barrier.

Despite the scientific breakthrough, there are concerns about whether lenacapavir will lead to equitable access. The USPSTF mandated health plans to cover PrEP and recommended coverage without cost-sharing [26, 27]. However, past experiences with oral PrEP, showed significant access disparities, especially for communities of color and cisgender women in the U.S. Prior authorization requirements, especially in geographic regions in the us which a high incidence of HIV impede the access to PrEP medications [28]. Such requirements in turn amplify the access disparities in terms of gender and race. The potential high cost of lenacapavir, raises concerns as its existing use as an anti-retroviral medication is priced at \$42,250 per year in the US[29]. This could deter health plans from covering the medication. Step therapy and prior authorization requirements by help plans

for long-acting injectable PrEP such as lenacapavir, need to be carefully monitored to ensure that the step therapy requirements reflect clinical guidelines accurately. There is a need for updated guidance for health plans from CMS to navigate coverage of PrEP equitably [30]. A high price for lenacapavir would burden public health programs and Medicaid with tight budgets, making it difficult to provide access to uninsured populations or the low-income population. In an effort to curb drug prices, Gilead plans to grant manufacturing licenses to six generic manufacturers to produce low cost lenacapavir to make it accessible to low and middle income countries with disproportionate HIV burden[31].

Another challenge in lenacapavir access is its effective integration into clinical practice, which will require concerted efforts, especially to educate and inform both healthcare providers and patients about this option. Programs to enhance healthcare provider's knowledge and awareness of the eligibility criteria, administration and potential side effects of lenacapavir. Preparedness in terms of infrastructure to store and prepare injections and training personnel to handle the same is crucial. Providers should also have familiarity regarding the reimbursement options for lenacapavir as PrEP and link eligible candidates to patient assistance programs if available. Additionally, patients may have concerns about injections and the injection site pain or reactions which should be openly addressed by the providers. Since long-acting PrEP requires fewer doses than daily oral PrEP, providers should emphasize the need for adherence to the injection schedule and proactively set up timely appointments and reminders to help patients stay on track. In these cases, collaborative care models with pharmacists have been demonstrated to be successful in PrEP delivery [32, 33] and can also be leveraged by providers to enable successful implementation of lenacapavir for PrEP.

Community pharmacies have great potential to increase uptake of long-acting PrEP as they are relatively easily accessible healthcare resources that can break down some of the barriers to PrEP[34]. Approximately 9 out of 10 people in the United States live within 5 miles of a pharmacy[35]. Recognizing the need for increased PrEP access, legislation has been passed or introduced enabling pharmacists to prescribe oral PrEP in 9 states[36]. Additionally pharmacists are allowed to prescribe injectable PrEP in only 4 states[36]. When a prescription for PrEP is available, only 36 of the 50 states allow a pharmacist to administer injectable PrEP[36]. Studies have shown that pharmacist-led PrEP programs that allow pharmacists to prescribe PrEP show promise for PrEP initiation and uptake and should be leveraged in the roll out of lenacapavir as PrEP[37].

However, to enable pharmacists to deliver lenacapavir for PrEP certain challenges need to be overcome. Barriers to initiating PrEP in the pharmacy setting exist, such as staffing, resources, and time required to screen patients for PrEP

eligibility (e.g., perform HIV testing), particularly because these services are time consuming, may be ineligible for reimbursement, and/or may not generate revenue[37, 38]. Not all states allow pharmacists to perform an HIV test which are mandatory before initiating PrEP[36]. Pharmacists face significant barriers in prescribing PrEP due to inadequately formulated policy regarding reimbursement for their services and policy that restricts their ability to perform lab tests[36]. In a recent update from the Centers for Medicare & Medicaid Services (CMS), FDA-approved PrEP using antiretroviral drugs and the administration of injectable PrEP are covered under Part B as a preventive service [39]. While formulating the regulation despite several calls from various organizations to consider the significant role of the pharmacists in PrEP service the CMS declared that pharmacists are not recognized Medicare providers or suppliers and were not eligible to be paid directly for furnished services [39]. Establishing legal frameworks to work around these would position pharmacist-initiated PrEP as key strategy for increasing uptake and enable equitable access to PrEP, especially lenacapavir.

Compared to existing literature, this review provides a consolidated and updated overview of lenacapavir as an option for long-acting PrEP. We summarized all clinical trials for lenacapavir as PrEP without being confined to strict inclusion/exclusion criteria pulling together diverse findings. We also including a concise overview of the trials planned in the future. Additionally, the strength of this narrative review lies in the fact that it draws attention to the practical challenges and considerations to be kept in mind for a successful roll-out of lenacapavir as PrEP. This review is limited by being a narrative and not a systematic literature review. The studies included are methodologically heterogeneous, and these are distinctive clinical trials being performed among different populations. Furthermore, clinical trials on the topic are ongoing. Despite these limitations, this review acts as a primer and a call to action for the equitable access to lenacapavir as PrEP.

Conclusion

Lenacapavir is a promising option for HIV PrEP due to its convenient dosing schedule, which can improve adherence, enhance privacy, and be accessible to individuals with limited healthcare access. It has shown good long-term efficacy and safety in preventing HIV, making it a valuable tool in HIV prevention efforts. However, ongoing research is needed to further confirm its long-term benefits across diverse populations. Addressing potential barriers, such as high costs and the need for regular HIV testing, will be essential to maximize its impact and help achieve the goal of ending the HIV epidemic by 2030.

Key References

- “PURPOSE Studies,” Gilead PrEP PURPOSE Studies. Accessed: Oct. 09, 2024. [Online]. Available: <https://www.purposestudies.com/>: This website was chosen as a key reference as it contains detailed information regarding each of the 5 planned PURPOSE studies that are assessing lenacapavir for preexposure prophylaxis (PrEP) across a diverse range of participants. The information on this website helps us keep abreast on the recent data and results from the ongoing PURPOSE programs.
- L.-G. Bekker et al., “Twice-Yearly Lenacapavir or Daily F/TAF for HIV Prevention in Cisgender Women,” *N. Engl. J. Med.*, vol. 0, no. 0, <https://doi.org/10.1056/NEJMoa2407001>. This publication summarizes the data from the pivotal Phase 3 clinical trial outlines the efficacy and safety of Lenacapavir for preexposure prophylaxis (PrEP) in cisgender women.
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Author Contribution S.G. conceptualized the idea. S.G., S.V. and P.M. wrote the main manuscript text and prepared Table 1. All authors reviewed the manuscript.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interest The authors declare no competing interests.

Human and Animal Rights and Informed Consent This article does not contain any studies with human or animal subjects performed by any of the authors.

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